

LESSON 7.6

MORE INTEGER ADDITION



LESSON INTRODUCTION

MA.6.NSO.4.1 Apply and extend previous understandings of operations with whole numbers to add and subtract integers with procedural fluency.

LEARNING TARGETS

- I can add integers with procedural fluency.

BENCHMARK COHERENCE

BUILDING ON

MA.5.NSO.2 Add, subtract, multiply, and divide multi-digit numbers.

WORKING TOWARD

MA.7.NSO.2.2 Add, subtract, multiply, and divide rational numbers with procedural fluency.

MA.7.NSO.2.3 Solve real-world problems involving any of the four operations with rational numbers.

ESSENTIAL QUESTIONS

- How do you add integers with opposite signs?
- How do you use absolute value when adding integers?

LESSON NARRATIVE

In this lesson, students practice adding integers to build their procedural fluency. Students play a game with playing cards to support them as they increase their fluency adding integers. Then they practice adding one-, two-, and three-digit integers.

COMPONENT SUMMARY

7.6.1 WARM-UP (~5 MIN.)

Write true addition equations using integers from -3 to 3

7.6.2 ACTIVITY (15 MIN.)

Play an integer addition game using playing cards

7.6.3 GUIDED PRACTICE (10 MIN.)

Partner or group work finding the sum of one-, two-, and three-digit integers

7.6.4 YOUR TURN! (10 MIN.)

Practice finding the sum of one-, two-, and three-digit integers

7.6.5 WRAP UP (~5 MIN.)

Demonstrate understanding of the lesson learning targets

RESOURCES

GLOSSARY

Key Terms: N/A

Additional Terms: N/A

MATERIALS

- One deck of cards per pair of students

ADDITIONAL PREPARATION

- Prepare an anchor chart with the card game rules.
- Prepare more difficult and less difficult problems for 7.6.3 Guided Practice and 7.6.4 Your Turn!.
- Prepare groupings of students for 7.6.2 Activity.

TEACHER TIPS

COMMON MISCONCEPTIONS

- Students may add integers of opposite signs and give the value a negative sign. For example, $-3+2\neq-5$.

THINGS TO CONSIDER

- Consider displaying an anchor chart with the card game rules for 7.6.2 Activity.
- The card game in 7.6.2 Activity will be more engaging for all students if students who work at similar rates are partners. Consider strategic grouping based on observations from the unit so far.

LITERACY AND LANGUAGE ROUTINES

Notice and Wonder

Consider implementing this routine before students complete question 2 in 7.6.3 Guided Practice. This will provide students the opportunity to consolidate their understanding of how to add signed integers.

MATHEMATICAL THINKING AND REASONING (MTR) STANDARD

MA.K12.MTR.5.1 Patterns and Structure

7.6.3 Guided Practice provides students an opportunity to look for similarities among integer addition problems. Ask assessing and advancing questions to guide consolidation of the rules for adding integers.

DIFFERENTIATE INSTRUCTION

Each activity in this lesson provides opportunity to differentiate instruction by content, by process, or by product.

- For 7.6.3 Guided Practice and 7.6.4 Your Turn!, consider substituting the given integer addition problems with less difficult problems if students are struggling. If students finish early, supplement with more complex problems, such as four-digit integers or sums with three addends.
- 7.6.2 Activity provides kinesthetic, auditory, and visual entry points as students play a card game.
- Consider allowing students to use words or visuals (such as positive and negative signs, integer chips) to represent their two tips in question 2 of 7.6.3 Guided Practice.

7.6.1 WARM-UP: TRUE EQUATIONS

Component Overview: Students fill in the blanks to write two addition equations using integers from -3 to 3.

QUESTIONING

For students who may finish early, consider having them create as many true equations as they can.

ENRICHMENT

MA.K12.MTR.3.1: Fluency

- Did you fill in the addends or the sum first?
- Which method made more sense for you? Why?



7.6.1 Warm-Up: True Equations

1. Using integers between -3 and 3 one time each, fill in the blanks to make the equation true.

Answers will vary.

$$\boxed{-3} + \boxed{2} = \boxed{-1}$$



7.6.2 Activity: Integer War

Set Up:

- Remove all jokers and face cards (Kings, Queens, and Jacks) from the deck.
- With your partner, determine who is the dealer. The dealer deals the entire deck of cards to each player, face down.

Rules:

- The black cards (Spades ♠ and Clubs ♣) represent positive values.
- The red cards (Diamonds ♦ and Hearts ♥) represent negative values.
- Aces represent either 1 or -1 , while the other cards represent the positive or negative value indicated on the card.

How to Play:

- To begin, each player will flip over one card from their deck and place it in the center.
- The first player to figure out the sum of the two integers keeps both cards and adds them to the bottom of their stack.
 - NOTE: If a player says an incorrect sum, the other player responds for a chance to win the hand before the first player can change their answer.
- Repeat this process until time is called or until one player has all of the cards in the deck.

7.6.2 ACTIVITY: INTEGER WAR

Component Overview: Students play a card game with a partner to practice adding integers. Consider displaying an anchor chart with the card game rules. The card game will be more engaging for all students if students who work at similar rates are partners.

DIFFERENTIATE INSTRUCTION

Consider your students and their unique learning strengths when planning this engaging activity.

- Model the game once as a whole group, making note of specific roles or tasks for students to complete per pairing.
- Shorten each game and make a tournament-style bracket in which winners of each pairing play each other and the losing partner must then become the winner's cheering section the rest of the activity.
- Split the class into two groups, with one group playing the game while the other engages with 7.6.3 Guided Practice, then switch.

QUESTIONING

Consider creating some groups of three to play the game where they are finding the sum of three integers.

ENRICHMENT

- How do you use the color of the cards to determine the sign of the sum?
- How could you use the symbols on the cards as counters to help you be sure of your sum?

7.6.3 GUIDED PRACTICE: FLUENCY WITH ADDING INTEGERS

Component Overview: Students practice finding the sum of one-, two-, and three-digit integers. The component concludes with students consolidating what they have learned by writing two tips for adding integers.

TEACHER MOVES

Notice and Wonder

Consider implementing this routine before students complete question 2. This will provide students the opportunity to consolidate their understanding of how to add signed integers.



7.6.3 Guided Practice: Fluency with Adding Integers

1. Find each sum.

- a. $-73 + 210 = 137$ b. $143 + (-9) = 134$
 c. $-18 + 875 = 857$ d. $-175 + (-544) = -719$
 e. $92 + (-100) = -8$ f. $-256 + (-301) = -557$

2. Write two tips to share with someone who is just learning how to add integers.

Answers will vary.

- If the integers have the same sign, add their absolute values. The sum will have the same sign as the integers.
- If the integers have opposite signs, subtract their absolute values. The sum will have the sign of the integer with the greater absolute value.

7.6.4 YOUR TURN!

Component Overview: Students continue to practice finding the sum of one-, two-, and three-digit integers.

ENRICHMENT

- How can you determine the sign of the sum when the integers have the same sign?
- How can you determine the sign of the sum when the integers have opposite signs?

DIFFERENTIATE INSTRUCTION

Consider displaying an anchor chart with instructions, visual models, and examples for adding integers with the same and with opposite signs. Referencing this chart will help students develop procedural fluency with adding integers.



7.6.4 Your Turn!

1. Find each sum.

- a. $10 + (-204) = -194$ b. $(-99) + (99) = 0$
 c. $904 + (-11) = 893$ d. $-1 + 76 = 75$
 e. $205 + (-643) = -438$ f. $-151 + (-70) = -221$
 g. $-399 + (-405) = -804$ h. $-80 + 606 = 526$
 i. $703 + (-190) = 513$ j. $-600 + (-78) = -678$

**7.6.5 Wrap-Up: Ticket Out the Door**

1. Find each sum.

a. $808 + (-547) = 261$

b. $-322 + (-65) = -387$

c. $-755 + 707 = -48$

7.6.5 WRAP-UP: TICKET OUT THE DOOR

Component Overview: Students demonstrate their understanding of adding integers with the same and opposite signs. Use data from this formative assessment to determine which students need remediation before the next lesson. Students should not use a calculator to complete the Wrap-Up.

QUESTIONING

To add complexity, ask students who finish early to write a real-world problem that could be modeled by the integer addition equations in the Wrap-Up.

Alternatively, challenge your students to play Integer War with a family member at home and report their activity.